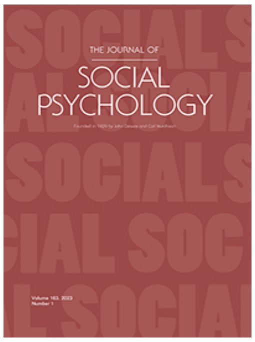

The Journal of Social Psychology

ISSN: (Print) (Online) Journal homepage: www.tandfonline.com/journals/vsoc20

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To cite this article: Qian Sun, Shasha Luo, Qianyun Gao, Wenjian Fan & Yongfang Liu (2023) Intuitive thinking impedes cooperation by decreasing cooperative expectations for pro-self but not for pro-social individuals, The Journal of Social Psychology, 163:1, 62-78, DOI: 10.1080/00224545.2022.2122768

To link to this article: https://doi.org/10.1080/00224545.2022.2122768

THE JOURNAL OF SOCIAL PSYCHOLOGY
2023, VOL. 163, NO. 1, 62–78
https://doi.org/10.1080/00224545.2022.2122768

Intuitive thinking impedes cooperation by decreasing cooperative expectations for pro-self but not for pro-social individuals

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ABSTRACT
This study conducted two experiments to explore the effects of intuitive thinking and social value orientation (SVO) on cooperative behavior and assess the mediating effect of cooperative expectations. It manipulated intuitive thinking by increasing the participants' need for cognitive closure, classified SVO using the triple-dominance measure, measured cooperative behavior using the prisoner's dilemma game, and considered cooperative expectations based on participants' assessments of the cooperativeness of their counterparts. Both experiments showed that intuitive thinking increased and decreased the cooperation of pro-social and pro-self individuals, respectively. In pro-self individuals, cooperative expectations mediated the effect of intuitive thinking on cooperation.

ARTICLE HISTORY
Received 15 September 2021
Accepted 1 September 2022

KEYWORDS
Cooperation; intuitive thinking; social value orientation; cooperative expectations

Introduction

Are people intuitively inclined to cooperate when facing social dilemmas, or are cooperation decisions the result of deliberative thinking? Recent studies have attempted to address this scientific question by focusing on the relationship between intuitive/deliberative thinking and cooperation (for reviews, see Bouwmeester et al., 2017; Rand, 2016). Some studies found that intuitive thinking promotes cooperation (e.g., Everett et al., 2017; Rand et al., 2012), whereas others failed to find such an effect (e.g., Tinghög et al., 2013). These inconsistent results may suggest that intuitive thinking does not always lead to cooperation, and other factors such as personality traits should be considered (for a similar proposition, see Rand & Kraft-Todd, 2014).

We focused on an important personality trait that may moderate the effect of intuitive thinking on cooperation in social dilemmas – social value orientation (SVO). Within the context of social dilemmas, people typically have to make decisions without knowing the decisions of others (Van Lange et al., 2013). Under such social uncertainty, cooperative expectations play a crucial role in determining people's ultimate cooperative behavior (Engel et al., 2021). Hence, we aimed to reveal the potential role of cooperative expectations in the relationships among intuitive thinking, SVO, and cooperative behavior.

Intuitive thinking and cooperation

Intuitive thinking refers to the rapid processing of information with little explicit deliberation (Kahneman, 2003; Rand, 2016). The intuitive model of prosociality proposes that cooperation stems from intuitive, reflexive, and even automatic processes (Zaki & Mitchell, 2013), thus suggesting that

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intuitive thinking promotes cooperation. Such a proposition can be found in studies that focused on the link between intuition and prosocial behavior that employed tactics to reduce cognitive control and the depth of information processing (e.g., time pressure, cognitive load, and need for cognitive closure) to induce intuitive thinking (Acar-Burkay et al., 2014; Everett et al., 2017; Rand et al., 2014). For instance, higher time pressure in the prisoner's dilemma game (PDG) would induce intuitive thinking and make people more willing to cooperate with others (e.g., Everett et al., 2017). Increasing cognitive load renders a higher need for cognitive closure (Halevy et al., 2006; Wu et al., 2020) and facilitates intuitive cooperation (e.g., Rand et al., 2014). Moreover, when faced with an increasing need for cognitive closure, people tend to make decisions faster (De Dreu et al., 1999; Roets et al., 2015; Schumpe et al., 2017) and behave more prosocially (e.g., Acar-Burkay et al., 2014).

However, the conclusions of certain studies are inconsistent with these findings (for reviews, see Kvarven et al., 2020). For instance, participants have been found to cooperate to the same degree with others in the PDG when making decisions with or without time pressure (Capraro & Cococcioni, 2015; Tinghög et al., 2013). A meta-analysis by Bouwmeester et al. (2017) revealed that inducing intuitive thinking has no effect on cooperation. Evidence from neuroscience has also demonstrated that the functions of the left dorsolateral prefrontal cortex of the brain are activated before making cooperative decisions, thereby revealing that cooperation requires exerting reflective control over selfish basic instincts (Steinbeis et al., 2012).

The lack of consensus in the literature indicates that the effect of intuitive thinking on cooperation is unstable and may be influenced by various factors. Cooperative behavior is the result of a combination of the decision-making context and decision-maker characteristics (Jagau & Veelen, 2017). When making cooperation decisions in a situation that promotes intuitive thinking, although the decision-maker performs the same decision-making task as they would in a situation that does not promote intuitive thinking, their thinking mode changes from deliberative to intuitive. This shift makes cooperation – a decision-making process with social overtones – largely contingent on the characteristics of the decision-maker themselves, and in particular, their social preferences. Therefore, we aimed to elaborate on what would be observed when examining the relationship between intuitive thinking and cooperation decisions, by taking social preferences into account.

SVO, intuitive thinking, and cooperation

The most typical characteristic that reflects the social preference of the decision maker is SVO. As a relatively stable personality trait (Hilbig et al., 2014; Hu & Mai, 2021; Van Lange, 1999), SVO refers specifically to a preference pattern that individuals manifest when allocating resources to the self and others in interdependent social contexts (Cornelissen et al., 2011; Van Lange et al., 1997). Researchers have identified three main types of SVO – prosocial (maximizing the joint outcomes for both parties or minimizing differences in outcomes between oneself and others), competitive (maximizing one's relative advantage over others), and individualism (maximizing one's own outcomes regardless of those of others) orientations (e.g., Van Lange et al., 1997). Generally, people of the first orientation are classified as pro-socials, whereas those of the latter two types are viewed as pro-selfs (Bogaert et al., 2008; De Dreu & Van Lange, 1995; Hu & Mai, 2021).

SVO is a critical factor that affects individual behavior and attitudes in a wide range of social situations. Numerous studies have found that pro-socials (vs. pro-selfs) are more willing to help the poor and the sick (e.g., Van Lange et al., 2007), are more attentive to fairness in outcomes for the self and others (Bieleke et al., 2017; De Dreu & Van Lange, 1995), and exhibit higher levels of cooperation in social dilemmas such as the PDG (Balliet et al., 2009; Fernandes et al., 2022; Pletzer et al., 2018). These findings indicate that pro-SVOs enhance pro-sociality and motivate individuals to engage in more cooperative behavior.

Moreover, intuitive cooperation differs in pro-social and pro-self individuals. Unlike pro-selfs, pro-socials tend to pay attention to others' outcomes and are interested in promoting the collective welfare (Joireman & Duell, 2005; Stouten et al., 2005). Hence, their intuitive response is to cooperate with others (Rand et al., 2012, 2014). Importantly, such behavior is not static, and might be altered after careful deliberation (Cornelissen et al., 2011). Indirect support for these notions can be derived from studies within the context of the dictator game, in which the allocator has the power to allocate a certain amount of money to the recipient. The results of these studies call into question the standard economic assumption that individuals act solely out of self-interest (Brañas-Garza et al., 2017). Cornelissen et al. (2011) found that pro-socials under high (vs. low) cognitive load allocate more money to the recipient, whereas pro-selfs allocate the same amount, regardless of the cognitive load condition. Moreover, Chen and Krajbich (2018) established that prosocial allocators choose a higher payoff for others in the high time pressure condition and a higher payoff for the self in the low time pressure condition, whereas selfish allocators do the opposite. In the context of the dictator game, the decision-maker influences the outcomes of the self and counterparts involved, but the other(s) have no power to impact the outcome of the decision-maker (Brañas-Garza et al., 2017). Thus, this game may not reflect real-life interactions, because social interactions are characterized by the interdependence of all counterparts (Rand, 2017). Therefore, we focused on social interactions within an outcome-interdependent context, in which each person can make a choice and their outcome would be determined by the joint choice.

Mischkowski and Glöckner (2016) used a correlational design that revealed how pro-socials cooperate less the longer their decision time, whereas pro-selfs consistently cooperate less regardless of the time they used to make decisions. In a similar vein, Yamagishi et al. (2017) observed a negative correlation between decision-making time and cooperation level in pro-socials. However, they also found a positive correlation between these two factors in pro-selfs, which was confirmed in a recent meta-analysis (Andrighetto et al., 2020). These findings accumulate insights into the relationships between SVO, intuitive thinking, and cooperation. However, rigorous causal moderating models have not been established. The abovementioned studies did not manipulate intuitive thinking and interpreted less decision time as the intuitive thinking mode. In fact, self-paced reaction time should be treated as a proxy for feelings of conflict rather than the degree of intuition versus deliberation (Rand, 2016). Therefore, we aimed to manipulate intuitive thinking directly and examine the moderating role of SVO in the effect of intuitive thinking on cooperative behavior.

Cooperative expectations, SVO, intuitive thinking, and cooperation

In social dilemmas within an outcome-interdependent context, people commonly need to make decisions without knowing their interaction partner's decision (Pletzer et al., 2018). In such settings, to avoid being exploited by non-cooperators and guide further behavior, people rely on their expectations of others' cooperativeness (Engel et al., 2021; Ng & Au, 2016). As proposed by goal-expectation theory (Pletzer et al., 2018; Pruitt & Kimmel, 1977), individuals will choose to cooperate if they expect others to cooperate as well. This notion is supported by multiple lines of evidence (Balliet & Van Lange, 2013; Cigarini et al., 2020; Engel et al., 2021; Ng & Au, 2016). For instance, Ng and Au (2016) measured cooperative expectations by asking participants to assess the likelihood of their interaction partner's cooperativeness, and reported that the higher cooperative expectations the participants hold, the more likely they are willing to cooperate with this interaction partner.

According to social projection theory (Kelley & Stahelski, 1970; Krueger, 2007) and structural assumed similarity bias (Aksoy & Weesie, 2012; Kuhlman et al., 1992), individuals expect others to have similar psychological and behavioral responses to themselves. If people's cooperative behavior changes owing to changes in their thinking mode, these changes might serve as an internal cue for inferring others' beliefs, attitudes, and behavior. In other words, intuitive thinking may also affect expectations of others' cooperativeness, and follow an influencing pattern similar to that for cooperative behavior. Acar-Burkay et al. (2014) found that decisions to trust friends vary in participants with different needs for cognitive closure, which suggests that individuals' positive (cooperative) expectations of others would change under a different thinking mode.

The intuitive thinking–cooperative expectations link may be heterogeneous between pro-socials and pro-selfs. Pro-socials may intuitively have positive pro-sociality expectations of others and expect others to engage in high levels of pro-social behavior (e.g., cooperative behaviors) that match their own (Declerck & Bogaert, 2008; Kelley & Stahelski, 1970). Meanwhile, pro-selfs may intuitively expect others to engage in high levels of self-interested behavior (Declerck & Bogaert, 2008; Kelley & Stahelski, 1970). Notably, the behavior pattern could change when people make decisions relying more on deliberation rather than intuition (Andrighetto et al., 2020; Krueger et al., 2012). If this is the case, expectations of others may change accordingly. Specifically, when the thinking mode shifts from high to low intuition, pro-socials may start to have lower expectations of others' cooperativeness, and pro-selfs may no longer have lower cooperative expectations.

Together with the path analyses discussed above, cooperative expectations may be a potential bridge to elucidate the moderating role of SVO in the relationship between intuition and cooperative behavior. Specifically, intuitive thinking promotes higher cooperative expectations, and consequently, higher cooperative behavior in pro-socials. For pro-selfs, intuitive thinking reduces their cooperative expectations, which further impedes their inclination to behave cooperatively.

The present study

We tested the hypothetical mediated moderation model (Figure 1) in two experiments. Experiment 1 examined the moderating role of SVO in the causal relationship between intuitive thinking and cooperative behavior, and Experiment 2 further explored the mediating role of cooperative expectations in the underlying process of Experiment 1. We induced participants' intuitive thinking with the need for cognitive closure task (Acar-Burkay et al., 2014) and classified them into pro-socials and pro-selfs using the triple-dominance measure (Van Lange et al., 1997). We studied cooperation within the context of the PDG, which is a typical outcome-interdependent paradigm that models the conflict between self-interest and collective welfare (Dreber et al., 2014; Rekosh & Feigenbaum, 1966) and elucidates individuals' expectations toward a partner's cooperativeness (Engel et al., 2021).

Experiment 1

Method

Design and participants

A power analysis indicated that at least 146 participants would be required to observe a statistically significant interactive effect in a two-way analysis of variance, assuming a medium effect size ($f = 0.25$) with sufficient power ($1–\beta > 0.85$) (using G*power 3.1; Faul et al., 2007). We recruited as many participants as possible in four weeks through advertising on the official recruitment platform of East China Normal University and ultimately recruited 207 college students (138 females) aged 18 to 29 ($M = 20.70$, $SD = 2.38$). They were randomly assigned to one of the two conditions (Intuitive thinking:

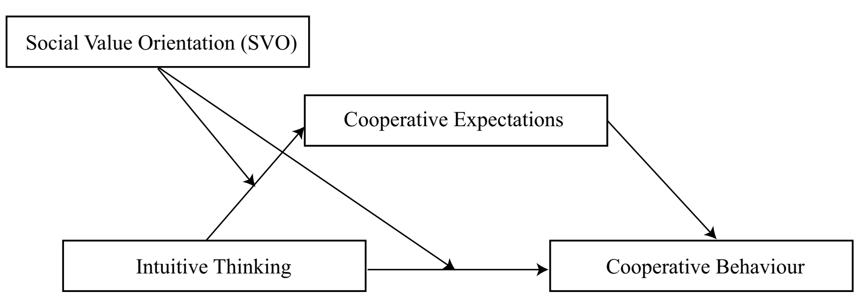

Figure 1. Conceptual mediated moderation model.

high/low intuition) after completing the SVO measurement. Cooperative behavior was measured using the PDG, in which two laboratory assistants (1 female, 1 male; 20 years old) were chosen to play the role of the participants' partner.

Materials and procedures

Brief instructions. As in a previous study (Sun et al., 2021), upon arrival to the laboratory, the participants were introduced to the two-person interaction experiment. They had a brief meeting with a same-sex stranger, with whom they would complete the experiment in different rooms and interact via a computer network. They were also informed that this experiment included a questionnaire task (i.e., SVO measurement) to identify their personality traits and a key-press task (i.e., cooperative behavior measurement; PDG) to elucidate interpersonal interactions. They also learned that they would receive a certain amount of money in each task. After the orientation, the Partner (i.e., the same-sex stranger) left and entered another room.

SVO measurement. We assessed SVO using the triple-dominance measure, which requires participants to make a choice from three alternative combinations of outcomes between the self and others (Declerck & Bogaert, 2008; Van Lange et al., 1997). The instructions for participants were as follows. Imagine you are randomly paired with another person. You two did not know each other, and would not meet in the future. You will be shown nine different matrices, and each matrix contains three options indicating outcomes of points for you and the other person. Both of you are required to choose one option to allocate points between the two of you. The more points one gets, the better outcome one will obtain. An example of a matrix is as follows. Option A, 560 points for the self and 300 points for the other (i.e., individualistic option); Option B, 500 points for the self and 500 points for the other (i.e., pro-social option); and Option C, 500 points for the self and 100 points for the other (i.e., competitive option). The payoff for participating in this task was fixed at 5 CNK (approximately 0.77 USD).

Intuitive thinking manipulation. We manipulated intuitive thinking using the need for cognitive closure task, in which a loud noise induces higher intuitive thinking (Kruglanski & Webster, 1991). Ambient noise has been widely used to increase the level of need for cognitive closure in prior research (e.g., Livi et al., 2015). Information processing would be effortful and difficult with a loud noise, and, hence, more psychologically costly. Thus, noise can induce intuitive thinking by increasing the level of need for cognitive closure (Acar-Burkay et al., 2014; Kruglanski & Webster, 1991). All the participants were asked to wear earphones until they completed the PDG. In the high-intuition condition, the audio in the earphones was sampled from a construction site. The volume was higher than 40 dB but without being uncomfortable, and its frequency ranged from 1.8 k to 10 k Hz. In the low-intuition condition, no audio was played in the earphones.

Cooperative behavior measurement. In this task, cooperative behavior was measured within the context of the PDG, in which two players independently decide whether to "cooperate" or "defect" and then obtain a certain amount of money based on their joint choice (Dreber et al., 2014; Rekosh & Feigenbaum, 1966). The PDG mimics the conflicts between collective welfare (i.e., by choosing to cooperate) and self-interest (i.e., by choosing to defect) using payoff matrices. Specifically, a combination of cooperation and defection would lead to the best outcome for the defector and the worst outcome for the cooperator; mutual defection would lead to a worse outcome for both individuals compared with mutual cooperation.

The participants performed 16 rounds of the PDG, each with slightly different payoff matrices and without feedback on each other's choice in between rounds. The options presented to participants were "A" or "B" rather than "cooperation" or "defection," to control the influence of semantic labels (Halevy et al., 2012). Additionally, the number of rounds was not presented explicitly to participants to control the end-of-game effect (Dreber et al., 2008). Cooperative behavior was calculated based on the ratio of rounds in which participants chose "A," which varied between 0 and 1.

An example matrix was shown to help the participants understand the rules of the task. The participants were informed that the payoff for participating in this task was determined by one round randomly selected by the computer. Before the formal round, we asked the participants to complete the comprehension item: "According to the example matrix, if you choose A and your Partner chooses B, how much will the two of you gain, respectively?" All the participants passed the comprehension test in one attempt and subsequently proceeded to the formal PDG.

Manipulation check. After the PDG, we instructed the participants to take off the earphones. To check the effectiveness of the intuitive thinking manipulation, we measured the participants' cognitive self-evaluation, reaction time (RT), and physiological arousal.

For the cognitive self-evaluation aspect, we asked the participants to report the extent to which they needed to make quick decisions during the decision-making process, and the extent to which they tended to simply process information, on a 15-point Likert scale ranging from 0 (not at all) to 14 (extremely; Webster, 1993). We also considered the RT of each round in the PDG as an index (Webster, 1993). We then assessed the participants' physiological arousal using the general activation scale (Thayer, 1967), which includes five items along the dimensions of full of pep, energetic, vigorous, active, and lively on a 10-point Likert scale ranging from 0 (not at all) to 9 (extremely). The measurement of physiological arousal was conducted to confirm that the loud noise only increased the tendency to process information intuitively rather than resulting in physiological arousal (Webster et al., 1996).

Final debriefing. Following the manipulation check questionnaire, the participants were paid 8.5 CNK (approximately 1.31 USD) on average. They were debriefed. During the debriefing, participants did not express any suspicion about the experimental setup and learned that the decisions made by their interaction partner were pre-programmed.

Results

The participants were considered to have a stable SVO if they consistently chose at least six items on the triple-dominance measure (e.g., Declerck & Bogaert, 2008; Van Lange et al., 1997). Based on this criterion, 196 participants (129 females; M_{age} = 20.67 years, SD = 2.30 years) were included in the data analysis. Of them, 99 ($N_{\text{pro-selfs}}$ = 44) belonged to the high-intuition condition and 97 ($N_{\text{pro-selfs}}$ = 44) belonged to the low-intuition condition. Table 1 presents the descriptive statistics for cooperative behavior, cognitive self-evaluation, RTs, and physiological arousal.

Manipulation checks

We conducted independent sample t-tests on cognitive self-evaluation, RT, and physiological arousal. The results revealed that participants in the high-intuition condition reported stronger needs to make a quick decision and process information in simpler ways compared with those in the low-intuition condition, $t(195) = -2.84$, $p = .005$, 95% CI [−1.46, − 0.26], Cohen's $d = 0.41$. Moreover, participants in the high-intuition condition spent less time, $t(195) = 3.03$, $p = .003$, 95% CI [0.75, 3.53], Cohen's d =

Table 1. Descriptive statistics of cooperative behavior, cognitive self-evaluation, RTs, and physiological arousal in Experiment 1.

Variables	High Intuition	Low Intuition
CB	0.62 (0.31)	0.61 (0.31)
CB (pro-socials)	0.78 (0.23)	0.65 (0.30)
CB (pro-selfs)	0.43 (0.28)	0.55 (0.31)
Cognitive self-evaluation	6.54 (2.02)	5.68 (2.22)
RTs	6.77 (4.03)	8.91 (5.74)
Physiological arousal	5.91 (1.77)	6.19 (1.48)

CB denotes cooperative rate. Mean scores are shown out of parentheses, and SDs are shown in parentheses. RTs denotes reaction times.

0.43. As for physiological arousal, we observed no significant difference for intuitive thinking, $t(195)$ = 1.18, p = .240, 95% CI [−0.19, 0.73], Cohen's d = 0.17. These results suggest that ambient noise effectively manipulated the tendency to make decisions relying more on intuition rather than physiological arousal; that is, intuitive thinking was effectively induced.

Moderation model test

We conducted the bootstrapping method for moderation (5,000 bootstrap samples, 95% confidence interval, model 1) of Hayes (2018) to test whether the effect of intuitive thinking (low intuition = – 1, high intuition = 1) on cooperative behavior (Z-Score) is moderated by SVO (pro-selfs = – 1, pro-socials = 1). The results (see Figure 2) indicated that (a) Model 1 was significant, $F(3,192) = 14.38$, $p <$ .001, $R^2 = .18$; (b) intuitive thinking failed to directly predict cooperative behavior, β = 0.001, SE = .07, 95% CI [−0.13, 0.13]; and (c) the interaction effect of intuitive thinking and SVO was significant, β = 0.21, $t(195) = 3.17$, SE = .07, 95% CI [0.08, 0.34]. Therefore, SVO moderated the relationship between intuitive thinking and cooperative behavior.

Simple slope analysis (Figure 3) further demonstrated that intuitive thinking positively predicted pro-socials' cooperative behavior (β = 0.21, SE = .10, 95% CI [0.04, 0.38]) but negatively predicted pro-selfs' cooperative behavior (β = – 0.21, SE = .09, 95% CI [−0.40, – 0.01]).

Discussion

Experiment 1 examined the moderating role of SVO in the causal relationship between intuitive thinking and cooperative behavior within an outcome-interdependent context. Our results established that the higher their intuitive thinking, the more cooperative behavior the pro-socials would adopt;

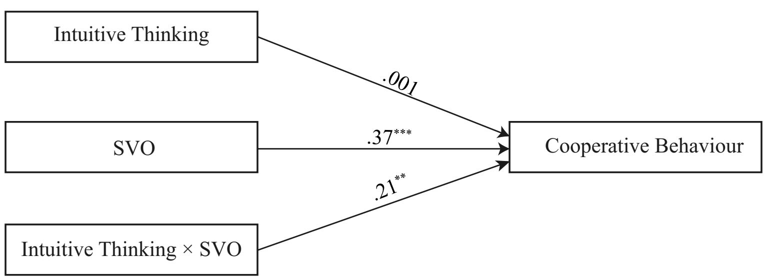

Figure 2. Moderation effect of social value orientation in the relationship between intuitive thinking and cooperative behavior. Note: SVO denotes social value orientation. Standardized values are presented.**p < .01, ***p < .001.

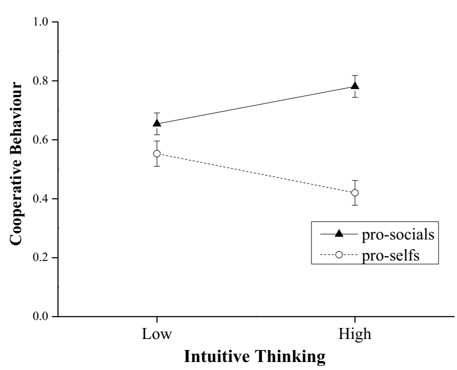

Figure 3. Interaction effect of intuition and social value orientation on cooperative behavior. Note: Error bars denote standard errors.

whereas the opposite applied to the pro-selfs. These findings suggest that whether intuitive thinking promotes cooperation depends on individuals' SVO.

Thus far, whether intuitive thinking directly affects cooperative expectations remains unclear. Research has noted that cooperation requires both individuals' own pro-social preference and their cooperative expectations toward others (Pruitt & Kimmel, 1977). Accordingly, Experiment 2 aimed to further explore the role of cooperative expectations in the moderating model of Experiment 1.

Experiment 2

Experiment 2 explored the role of cooperative expectations in the relationships among intuitive thinking, SVO, and cooperative behavior. We proposed that pro-socials would have higher cooperative expectations of their partners and increased cooperative behavior in social dilemmas when relying more on intuition, while pro-selfs would have lower cooperative expectations and decreased cooperative behavior when relying more on intuition.

Method

Design and participants

The sample size determination was identical to that of Experiment 1. We recruited participants via social media networks (WeChat) to increase the diversity of the sample, and thus, test the generalizability of the results of Experiment 1. We eventually recruited 184 participants (94 females; M_{age} = 24.33 years, SD = 5.20 years, range: 18 to 42 years), with 75 undergraduates and 109 employed people. We randomly assigned participants to either the high- or low-intuition condition after completing the online SVO measurement. Cooperative behavior was measured using an online version of the PDG.

Materials and procedures

Brief instructions. The instructions were similar to those in Experiment 1, except that the participants in this experiment were informed that they would be interacting with a same-sex stranger randomly matched online as their partner.

SVO measurement. Both the materials and procedures were identical to those of Experiment 1.

Intuitive thinking manipulation. Before the PDG, intuitive thinking was manipulated according to the procedure in Experiment 1.

Cooperative behavior measurement. Cooperative behavior was measured within the context of the PDG. The materials used were identical to those in Experiment 1, but here the implications of the two options were balanced to control individuals' letter preferences (i.e., A or B). All the participants were informed to choose "A" or "B" when one payoff matrix was presented. For half of the participants, option A referred to "cooperation" and option B referred to "defection," whereas for the other half, the implications were reversed. The procedures were identical to those in Experiment 1, but here the number of rounds was decreased to five. This modification was intended to ease the burden of repetition of too many rounds (Faust et al., 2018).

Manipulation check. As in Experiment 1, we measured the participants' cognitive self-evaluation, RT, and physiological arousal. For cognitive self-evaluation, the participants assessed the extent to which they had the need to make a slow decision and think longer (Webster et al., 1996). For physiological arousal, the scale ranged from 1, given that the arousal level of a living person cannot be 0.

Cooperative expectations measurement. Measuring cooperative expectations before cooperative behavior might increase the participants' perspective taking and lead to more cooperation (Galinsky et al., 2008). Thus, we measured the participants' cooperative expectations after five rounds of the PDG. We asked them to report how often they expected their partner to choose the cooperative option on a five-point Likert scale ranging from 1 (not at all) to 5 (extremely; Ng & Au, 2016; Van Lange, 1992).

Final debriefing. Following the manipulation check questionnaire, the participants were paid 8.2 CNK (approximately 1.27 USD) on average, and then debriefed.

Results

After excluding participants with unstable SVO, 169 participants (84 females; M_{age} = 24.44 years, SD = 5.29 years) were included in the data analysis, with 68 undergraduates and 101 employed people. Of them, 83 ($N_{\text{pro-selfs}}$ = 42) belonged to the high-intuition condition and 86 ($N_{\text{pro-selfs}}$ = 43) belonged to the low-intuition condition. Table 2 presents the descriptive statistics for cooperative behavior, cooperative expectations, and the manipulation checks of cognitive self-evaluation, RTs, and physiological arousal.

Manipulation checks

The results of the independent sample t-test revealed that participants in the high-intuition condition reported a weaker need to make a slow decision and less thought in decision-making than those in the low-intuition condition, $t(168) = 2.55$, $p = .012$, 95% CI [0.22, 1.71], Cohen's $d = 0.39$. Moreover, participants in the high-intuition condition spent less time making decisions in the PDG, $t(168) = 3.77$, $p < .001$, 95% CI [0.87, 2.79], Cohen's $d = 0.58$. As for physiological arousal, we observed no significant difference in intuitive thinking, $t(168) = -0.27$, $p = .785$, 95% CI [−0.42, 0.32], Cohen's $d = 0.04$. Thus, Experiment 2 effectively induced intuitive thinking using ambient noise.

Moderation model test

Moderating role of SVO in the intuitive thinking–cooperative behavior link. The interaction effect of intuitive thinking and SVO was significant, $\beta = 0.23$, $t(168) = 3.30$, $p = .001$, 95% CI [0.09, 0.36], thus suggesting that SVO moderated the effect of intuitive thinking on cooperative behavior. The simple slope analysis (Figure 4a) further demonstrated that intuitive thinking positively predicted pro-socials' cooperative behavior ($\beta = 0.26$, $SE = .10$, 95% CI [0.06, 0.45] but negatively predicted pro-selfs' cooperative behavior ($\beta = -0.19$, $SE = .10$, 95% CI [−0.38, − 0.004].

Table 2. Descriptive statistics of cooperative behavior, cooperative expectations, cognitive self-evaluation, RTs, and physiological arousal in Experiment 2.

Variables	High Intuition	Low Intuition
CB	0.60 (0.41)	0.58 (0.36)
CB (pro-socials)	0.85 (0.25)	0.65 (0.33)
CB (pro-selfs)	0.35 (0.39)	0.50 (0.38)
CE	3.00 (1.40)	3.12 (1.31)
CE (pro-socials)	3.80 (0.95)	3.44 (1.26)
CE (pro-selfs)	2.21 (1.32)	2.79 (1.30)
Cognitive self-evaluation	8.13 (2.55)	9.09 (2.36)
RTs	9.26 (3.40)	11.09 (2.91)
Physiological arousal	5.46 (1.30)	5.41 (1.12)

CB denotes cooperative rate. CE denotes cooperative expectations. Mean scores are shown out of parentheses, and SDs are shown in parentheses. RTs denotes reaction times.

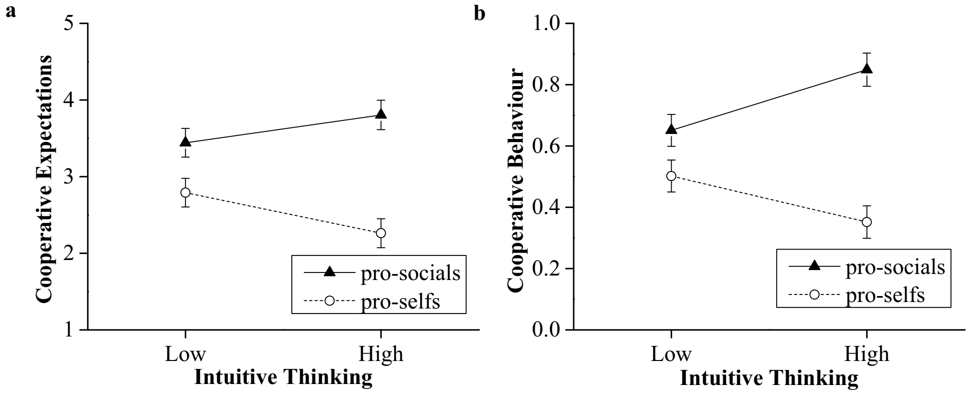

Figure 4. Interaction effect of intuition and social value orientation on cooperative expectations (a) and cooperative behavior (b). Note: Error bars denote standard errors.

Moderating role of SVO in the intuitive thinking–cooperative expectations link. The interaction effect of intuitive thinking and SVO was significant, β = 0.17, $t(168) = 2.50$, $p = .01$, 95% CI [0.04, 0.31], thus suggesting that SVO moderated the effect of intuitive thinking on cooperative expectations. The simple slope analysis (Figure 4b) further revealed that intuitive thinking negatively predicted pro-selfs' cooperative expectations (β = – 0.21, *SE* = .10, 95% CI [–0.41, – 0.02]) but did not significantly predict pro-socials' cooperative expectations (β = 0.13, *SE* = .10, 95% CI [–0.06, 0.33]).

Mediated moderation model test

We again used the bootstrapping method for moderation (5,000 bootstrap samples, 95% CI, model 8; Hayes, 2018) to test whether the mediating effect of cooperative expectations (Z-score) on the relationship between intuitive thinking (low intuition = – 1, high intuition = 1) and cooperative behavior (Z-score) would be moderated by SVO (pro-selfs = – 1, pro-socials = 1). The results (Figure 5) established that (a) Model 8 was significant, $F(3, 165) = 13.94$, $p < .001$, $R^2 = .20$; (b) the interaction effects of intuitive thinking and SVO on both cooperative behavior and cooperative expectations were significant (cooperative behavior: β = 0.09, *SE* = 0.04, 95% CI [0.01, 0.19]; and cooperative expectations: β = 0.17, *SE* = 0.07, 95% CI [0.04, 0.31]), thereby suggesting that SVO moderated not only the direct prediction of intuitive thinking on cooperative behavior but also the prediction of intuitive thinking on cooperative expectations.

For pro-socials, the mediating effect of cooperative expectations was insignificant, β = 0.10, Boot*SE* = 0.07, 95% CI [–0.03, 0.25]; and for pro-selfs, the mediating effect of cooperative expectations was significant, β = – 0.17, Boot*SE* = 0.08, 95% CI [–0.32, – 0.01]. Thus, cooperative expectations

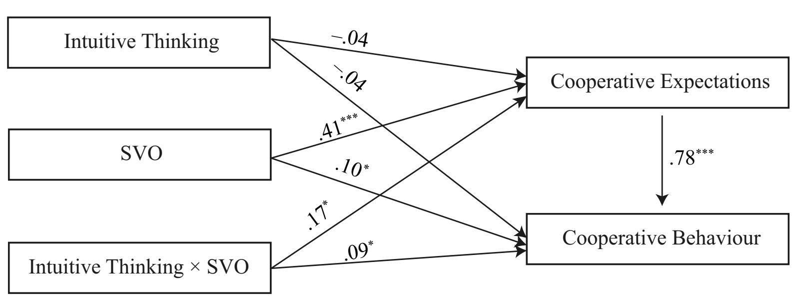

Figure 5. Mediated moderation model. Note: SVO denotes social value orientation. Standardized values are presented. $^{**}p < .01$, $^{***}p < .001$.

mediated the relationship between intuitive thinking and cooperation for pro-selfs but not for pro-socials.

Discussion

Experiment 2 confirmed that pro-socials chose more cooperative options in the high (vs. low) intuition condition, whereas pro-selfs did the opposite. These findings replicated the results of Experiment 1, thereby affirming that SVO played a significant moderating role in the relationship between intuitive thinking and cooperation. Moreover, the mediating effect of cooperative expectations was moderated by SVO. Specifically, intuitive thinking impeded cooperation by decreasing cooperative expectations for pro-selfs, whereas it directly promoted cooperative behavior without affecting cooperative expectations for pro-socials.

General discussion

Within an outcome-interdependent context, we explored the boundary (Experiment 1) and mechanism (Experiment 2) of the relationship between intuitive thinking and cooperation. The two experiments demonstrated that people's intuitive tendency to cooperate depends on their SVO. The moderating effect of SVO in the intuition–cooperation link was mediated by cooperative expectations (Experiment 2).

By taking SVO into account, this study elucidated the intuitive thinking–cooperative behavior link, thereby integrating the seemingly contradictory results in previous research. Intuitive thinking does not always make people more cooperative; it only does so in pro-social people. For pro-selfs, intuitive thinking does not promote but instead hampers the occurrence of cooperation. These findings suggest that lower intuitive thinking might reduce tendencies to act in line with SVO. Intuitive thinking simplifies information processing, and makes people more dependent on their own social preference and respond intuitively. The lack of intuitive thinking thus increases the complexity of information processing; makes them less dependent on their own social preference; and controls or inhibits intuitive responding, even making people behave in the opposite direction. This view has been partially confirmed by relevant research. For instance, Cornelissen et al. (2011) found that pro-socials become less prosocial when they are required to calculate benefits and costs in a more refined way and to weigh collective and personal interests in an integrated way. These findings provided experimental evidence for a recent meta-analysis (Andrighetto et al., 2020) and shed light on the crucial role of SVO in intuitive cooperation. They also matched prior findings within the outcome non-interdependent context (e.g., Chen & Krajbich, 2018). Meanwhile, our results on the intuitive thinking–cooperative behavior link for pro-selfs were not entirely consistent with previous findings (e.g., Cornelissen et al., 2011). This fact may be attributed, in part, to differences in the social interaction context and paradigm choices. Future studies could investigate these possibilities.

Remarkably, SVO moderated not only the intuitive thinking–cooperative behavior link but also the intuitive thinking–cooperative expectations link. For pro-selfs, their cooperative expectations decreased when their thinking mode changed from low to high intuition. For pro-socials, their cooperative expectations remained the same. Such a moderating pattern of SVO was not identical to that in the intuitive thinking–cooperative behavior link, which is a discrepancy with our initial hypothesis. This pattern suggests that cooperative expectations of pro-socials are relatively stable and might not switch across extremes with the changes in their thinking mode. Meanwhile, the cooperative expectations of pro-self are vulnerable to their thinking mode – intuitive cooperative expectations of pro-socials, rather than pro-selfs, are less affected by deliberative thinking (Krueger, 2007). These findings reveal that SVO is an important cognitive cue for expecting other's cooperativeness in social dilemmas with social uncertainty, which also lends support to social projection theory (Kelley & Stahelski, 1970; Krueger, 2007) and structural assumed similarity bias (Aksoy & Weesie, 2012; Kuhlman et al., 1992) to some degree.

The mechanism of intuitive thinking and cooperative behavior is heterogeneous for people with different SVOs. Specifically, intuitive thinking directly promotes pro-socials' cooperative behavior without affecting cooperative expectations, whereas it impedes pro-selfs' cooperative behavior by decreasing cooperative expectations. This difference may be due to the prosocial motivations of pro-socials and pro-selfs. Pro-socials tend to pursue mutual cooperation to maximize the collective interests and hold higher positive expectations toward others, even if their intuitive thinking is promoted (Pletzer et al., 2018), which may further mask the mediating role of cooperative expectations in the intuitive thinking–cooperative behavior link. Other mechanisms (e.g., perceptions of interpersonal closeness) might be implicated (e.g., Cornelissen et al., 2011), which could be explored by future studies.

For pro-selfs, such a mediating mechanism, at first glance, seems contrary to the fact that they may exploit partners with whom they are expected to cooperate to maximize self-interest (e.g., Boone et al., 2010). Although exploiting others could benefit pro-selfs, they may also gain from cooperation, such as creating a good reputation (Jordan & Rand, 2020) and avoiding negative emotions (e.g., compassion and guilt; Goetz et al., 2010). When making decisions relying less on intuition, individuals will analyze the possible behavior patterns of others more thoroughly. Specifically, in our experiment, pro-selfs would recognize that they will lose 0.50 CNK when choosing cooperation (vs. defection), but such a choice will bring them a good reputation and spare them from experiencing negative emotions, such as guilt. Thus, they may make more cooperative choices. Similarly, Pletzer et al. (2018) found that pro-selfs will not exploit a cooperative partner in general. However, it does not mean that pro-selfs will fully turn to cooperation after deliberation. Our data showed that both cooperation expectations and cooperative behavior of pro-selfs were lower than those of pro-socials. The facilitative effect of cooperative expectations on pro-selfs' cooperative behavior was derived by comparing the high- and low-intuition conditions. Pro-selfs overestimate self-interest as the main motivation of social interaction (Vuolevi & Van Lange, 2010) and are more likely to intuitively expect others to be less cooperative (Iedema & Poppe, 1994), which, in turn, guides them to choose non-cooperation to avoid exploitation by others. Once the thinking mode turns to deliberation, pro-selfs begin to eliminate their bias toward others' cooperativeness and increase their cooperative expectations toward others appropriately, which may further motivate them to become more cooperative. These findings indicate that pro-selfs can be encouraged to cooperate by reinforcing their cooperative expectations. They may maximize their own long-term outcomes by forming mutually beneficial cooperative relationships (Pletzer et al., 2018).

Our work addressed a very important and complex problem: how do simple (intuitive) and complex (deliberative) thinking affect cooperative behavior differently among people with diverse SVOs? Existing studies have emphasized the beneficial effect of intuitive thinking on cooperation, while ignoring the beneficial effect of deliberative thinking on cooperation. Human cooperative behaviors, especially the more complex ones, also evolved with the continuous evolution of human cognitive ability or rationality. Deliberative thinking does not always undermine cooperation; it might motivate cooperation by promoting strong reciprocity preferences and preventing selfish intuitions (e.g., Kocher et al., 2017). In a sense, the modern educational system is designed to make people more rational and cooperative. As with the effect of intuitive thinking on cooperation, whether deliberation will prevent cooperation should also depend on people's SVO. Given that intuitive thinking is not a normal and human-controlled processing mode, whereas the deliberative thinking mode is, exploring whether deliberative thinking makes people more uncooperative and who is more uncooperative seems to have high practical value and significance. Future research could explore this issue in a more general sense of the interaction between rationality and human nature.

Some limitations of this study should be noted. First, we explored the constructed mediated moderation model within a same-sex interaction context. However, cooperative behavior toward a same-sex person may differ from that toward an opposite-sex person (Sheppard et al., 2020). Our findings may not be applicable to situations in which people interact with an opposite-sex person. Second, the use of pre-programmed decisions by the interaction partner may be deceptive. Future research should use deceptive techniques with caution and conduct research in realistic two-person interaction contexts to fit APA ethical standards. Third, we used a single measure of SVO. Future studies could use a more sensitive and higher-resolution measurement (e.g., a slider measure, Murphy et al., 2011) to validate our findings. Finally, the relative measurement sequence of cooperative expectations and cooperative behavior in our work may control the possible effect of perspective taking to some extent but may also pose problems in which people rationalize their cooperative expectations after they have made their decision. However, considering that the relationship between cooperative expectations and behavior was not moderated by the relative measurement sequence (for a review, see Balliet & Van Lange, 2013) and cooperative expectations have strong and robust effects on cooperative behavior in social dilemmas (e.g., Lanzetta & Englis, 1989), cooperative expectations may not be merely a post hoc rationalization of one's own behavior. In fact, expectations may precede and determine cooperative behavior. Future studies could manipulate cooperative expectations to establish a causal inference for our models.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Funding

This work was supported by the China Scholarship Council [202006140079]; Research Project of Shanghai Science and Technology Commission [20dz2260300] and the Fundamental Research Funds for the Central Universities; National Social Science Foundation of China [15ZDB121].

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Ethical statement

This study was conducted in accordance with institutional and general ethical guidelines, and was approved by the Ethics Review Board of East China Normal University (HR384-2019).

Data availability statement

The data described in this article are openly available in the Open Science Framework at https://osf.io/75gwd.

Open Scholarship

This article has earned the Center for Open Science badges for Open Data and Open Materials through Open Practices Disclosure. The data and materials are openly accessible at https://osf.io/75gwd.

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